

12 General Renormalization Theory

12.1 The Floating Cutoff¹

Before closing this chapter, it is worth commenting on the relation between conventional renormalization theory and an approach pioneered by Wilson [18]. In Wilson’s method one imposes a “floating” finite ultraviolet cutoff (either sharp or smooth) at momenta with components of order Λ , and instead of taking $\Lambda \rightarrow \infty$, one requires that the bare constants of the theory (those appearing in the Lagrangian) depend on Λ in such a way that all observable quantities are Λ -independent.

It is convenient to work with dimensionless parameters. If a bare coupling or mass parameter $g_i(\Lambda)$ has dimensionality $[\text{mass}]^{\Delta_i}$, we define the corresponding dimensionless parameter $\mathcal{G}_i$ by

$$\mathcal{G}_i(\Lambda) \equiv \Lambda^{-\Delta_i} g_i(\Lambda). \quad (12.4.1)$$

Ordinary dimensional analysis tells us that the value of $\mathcal{G}_i$ at one value Λ' of the cutoff can be expressed as a function of the values of the $\mathcal{G}_j$ at another value Λ of the cutoff, and the dimensionless ratio Λ'/Λ :

$$\mathcal{G}_i(\Lambda') = F_i(\mathcal{G}(\Lambda), \Lambda'/\Lambda). \quad (12.4.2)$$

No dimensional parameters other than Λ' and Λ can appear in F , because no ultraviolet or infrared divergences can enter here; the difference between the constants at Λ and at Λ' arises from diagrams whose internal lines are restricted to have momenta between Λ and Λ' . Differentiating Eq. (12.4.2) with respect to Λ' and then setting Λ' equal to Λ yields a differential equation for $\mathcal{G}_i$:

$$\Lambda \frac{d}{d\Lambda} \mathcal{G}_i(\Lambda) = \beta_i(\mathcal{G}(\Lambda)), \quad (12.4.3)$$

where $\beta_i(\mathcal{G}) \equiv [\partial F_i(\mathcal{G}, z)/\partial z]_{z=1}$. The functions $\beta_i(\mathcal{G})$ may be calculated for small couplings in perturbation theory. This is Wilson’s version of the “renormalization group” equation, which will be discussed in somewhat different terms in Volume II.

The Lagrangian for any finite value of the cutoff defines an effective field theory, in which instead of (or in addition to) integrating out “heavy” particles like the electron in the work of Euler *et al.*, one integrates out *all* particles with momenta greater than Λ . Even if one starts with a theory with a finite number of coupling parameters $\mathcal{G}_i^0$ at some cutoff Λ_0 , at any other value of the cutoff the differential equation (12.4.3) will generally yield non-zero values for all couplings allowed by symmetry principles.²

We now distinguish between the renormalizable and unrenormalizable couplings, labelled $\mathcal{G}_a$ and $\mathcal{G}_n$, respectively, with a running over the finite number N of couplings (including masses) for which $\Delta_a \geq 0$, and n running over the infinite number of couplings with dimensionalities $\Delta_n < 0$. We want to show that if the couplings $\mathcal{G}_a(\Lambda_0)$ and $\mathcal{G}_n(\Lambda_0)$ at some initial cutoff value Λ_0 lie on a generic N -dimensional initial surface $\mathcal{S}_0$, then (with some qualifications) for $\Lambda \ll \Lambda_0$ they will approach a fixed surface $\mathcal{S}$ that is independent of both

¹This section lies somewhat out of the book’s main line of development, and may be omitted in a first reading.

²The only known exceptions to this rule are in theories based on supersymmetry [4].

Λ_0 and the initial surface.³ This fixed surface is stable, in the sense that from any point on the surface, the trajectory generated by Eq. (12.4.3) stays on the surface. Such a stable surface defines a finite-parameter set of theories whose physical content is cutoff-independent, which as argued in the previous section, is the essential property of renormalizable theories. Furthermore, this construction shows that a generic theory defined with cutoff Λ_0 will look for $\Lambda \ll \Lambda_0$ like a renormalizable theory.⁴

To prove these results, consider any small perturbation $\delta\mathcal{G}_i(\Lambda)$ in the values of the $\mathcal{G}_i(\Lambda)$ satisfying Eq. (12.4.3). It will satisfy the differential equation

$$\Lambda \frac{d}{d\Lambda} \delta\mathcal{G}_i(\Lambda) = \sum_j M_{ij}(\mathcal{G}(\Lambda)) \delta\mathcal{G}_j(\Lambda), \quad (12.4.4)$$

where

$$M_{ij}(\mathcal{G}) \equiv \frac{\partial}{\partial \mathcal{G}_j} \beta_i(\mathcal{G}). \quad (12.4.5)$$

This equation couples the renormalizable and unrenormalizable couplings, making it difficult to see the difference in their behavior. To decouple them, we introduce the linear combinations

$$\xi_n \equiv \delta\mathcal{G}_n - \sum_{ab} \frac{\partial \mathcal{G}_n}{\partial \mathcal{G}_a^0} \left(\frac{\partial \mathcal{G}}{\partial \mathcal{G}^0} \right)_{ab}^{-1} \delta\mathcal{G}_b, \quad (12.4.6)$$

where $\mathcal{G}_a^0$ are the values of the renormalizable couplings at cutoff Λ_0 , which we shall use as coordinates for the initial surface, and $\mathcal{G}_n$ are the values of the non-renormalizable couplings at cutoff Λ derived from the differential equation (12.4.3), with initial value for Λ_0 at the point on the initial surface with coordinates $\mathcal{G}_a^0$. To calculate the derivative of ξ_n with respect to Λ , we note that the derivatives $\partial\mathcal{G}_i/\partial\mathcal{G}_a^0$ satisfy the same differential equation (12.4.4) as the $\delta\mathcal{G}_i$. It is an elementary exercise then to show that

$$\Lambda \frac{d}{d\Lambda} \xi_n = \sum_m N_{nm} \xi_m, \quad (12.4.7)$$

where

$$N_{nm} \equiv M_{nm} - \sum_{ab} \frac{\partial \mathcal{G}_n}{\partial \mathcal{G}_a^0} \left(\frac{\partial \mathcal{G}}{\partial \mathcal{G}^0} \right)_{ab}^{-1} M_{bm}. \quad (12.4.8)$$

Now we must estimate the elements of N_{nm} . For a free-field theory no cutoff is needed, so for very small coupling all bare parameters $g_i(\Lambda)$ become Λ -independent. Hence for small couplings the dimensionless parameters $\mathcal{G}_i$ simply scale as $\Lambda^{-\Delta_i}$, and the matrix M_{ij} is given by

$$M_{ij} \approx -\Delta_i \delta_{ij}. \quad (12.4.9)$$

³This theorem is due to Polchinski [19]. What follows here is a shortened and less rigorous version. (In Polchinski's proof, the initial surface is taken to be that with all non-renormalizable couplings vanishing. As we shall see here, the couplings approach the same fixed surface for generic initial surfaces.)

⁴Of course some theories have symmetries and a field content that do not allow any renormalizable interactions. This is the case for theories containing only fermion fields, or only the gravitational field. Such theories for $\Lambda \ll \Lambda_0$ look like free-field theories.

It follows that the matrix N_{nm} is given approximately by $-\Delta_n \delta_{nm}$. The defining characteristic of the non-renormalizable couplings is that $\Delta_n < 0$, so Eq. (12.4.7) tells us that, at least for couplings in some finite range, where N_{nm} is positive-definite, the ξ_n decay for $\Lambda \ll \Lambda_0$ like positive powers of Λ/Λ_0 . In this limit, then, the perturbations are related by

$$\delta \mathcal{G}_n = \sum_{ab} \frac{\partial \mathcal{G}_n}{\partial \mathcal{G}_a^0} \left(\frac{\partial \mathcal{G}}{\partial \mathcal{G}^0} \right)_{ab}^{-1} \delta \mathcal{G}_b. \quad (12.4.10)$$

In particular, if we make a small change in the initial surface $\mathcal{S}_0$ and/or the starting point on that surface and/or the initial cutoff Λ_0 , such that the perturbations $\delta \mathcal{G}_a$ in the renormalizable couplings vanish at some cutoff $\Lambda \ll \Lambda_0$, then the perturbations $\delta \mathcal{G}_n$ in all the other couplings at cutoff Λ also vanish. Thus the non-renormalizable couplings $\mathcal{G}_n(\Lambda)$ for $\Lambda \ll \Lambda_0$ can depend only on the renormalizable couplings $\mathcal{G}_a(\Lambda)$, not separately on the initial surface or the starting point on that surface or the initial cutoff Λ_0 . At cutoff $\Lambda \ll \Lambda_0$ all the couplings therefore approach an N -dimensional surface $\mathcal{S}$, with coordinates $\mathcal{G}_a(\Lambda)$, which is independent of both the initial surface and of Λ_0 . Note that the non-renormalizable couplings $\mathcal{G}_n$ are not generally small on $\mathcal{S}$; the important point is that they become functions of the renormalizable couplings. Changes in Λ with Λ remaining much less than Λ_0 will change the couplings, but the couplings will remain close to $\mathcal{S}$ (at least as long as the couplings do not become so large that N_{nm} is no longer a positive-definite matrix). Hence $\mathcal{S}$ is a stable surface, as was to be proved.

We have seen that all physical quantities may be expressed in terms of Λ and the $\mathcal{G}_n(\Lambda)$, and are Λ -independent. This is true in particular of the N conventional renormalized couplings and masses, like e and m_e in quantum electrodynamics. But we can then invert this relation, and express the $\mathcal{G}_n(\Lambda)$ in terms of the conventional parameters and Λ . In this way we can justify the usual renormalization program; all physical quantities are expressed in a cutoff-independent way in terms of the conventional renormalized couplings and masses.

The Wilson approach has some advantages in practice. One does not have to worry about subintegrations and overlapping divergences; the momentum cutoff applies to all internal lines. Also, some of the non-renormalization theorems of supersymmetry theory, which tell us that certain couplings are not affected by radiative corrections, work only for the cutoff-dependent bare couplings [20].

On the other hand, there are disadvantages to the Wilson approach. One must give up the special simplicities of working with renormalizable theories like quantum electrodynamics; once one starts integrating out particles with momenta above some scale Λ , the resulting effective field theory will contain all Lorentz- and gauge-invariant interactions, with Λ -dependent couplings. (Nevertheless, in physical processes at energies $E \ll \Lambda$, the dominant couplings will still be the renormalizable ones.) Also, the cutoff generally destroys *manifest* gauge invariance, and either manifest Lorentz invariance or unitarity. None of this is a problem in condensed matter physics, the original context of Wilson's approach, because no one would expect a realistic condensed matter theory to be strictly renormalizable, and there are no fundamental physical principles that are necessarily violated by a cutoff. In fact, in crystals there is a cutoff on phonon momenta, provided by the inverse lattice spacing.

At bottom, the difference between the conventional and the Wilson approach is one of mathematical convenience rather than of physical interpretation. Indeed, conventional renormalization already provides a sort of adjustable cutoff; when we express our answer in terms of coupling constants that are defined as the values of physical amplitudes at some momenta of order μ (as for the scalar field theory discussed in the previous section), the cancellations that make integrals converge begin to operate at virtual momenta of order μ . Conversely, the Λ -dependent coupling constants of the Wilson approach must ultimately be expressed in terms of observable masses and charges, and when this is done the results are, of course, the same as those obtained by conventional means.

12.2 Accidental Symmetries⁵

In Section 12.3 we saw that there are good reasons to adopt renormalizable field theories as approximate descriptions of nature at sufficiently low energy. It often happens that condition of renormalizability is so stringent that the effective Lagrangian automatically obeys one or more symmetries, which are not symmetries of the underlying theory, and may therefore be violated by the suppressed non-renormalizable terms in the effective Lagrangian. Indeed, most of the experimentally discovered symmetries of elementary particle physics are “accidental symmetries” of this sort.

A classic example is provided by the inversions and flavor conservation in the electrodynamics of charged leptons. The most general renormalizable and gauge- and Lorentz-invariant Lagrangian density for photons and fields ψ_i of spin $\frac{1}{2}$ and charge $-e$ takes the form

$$\begin{aligned}\mathcal{L} = & -\frac{1}{4}Z_3F_{\mu\nu}F^{\mu\nu} \\ & -\sum_{ij}Z_{Lij}\bar{\psi}_{Li}[\not{\partial} + ie\not{A}]\psi_{Lj} - \sum_{ij}Z_{Rij}\bar{\psi}_{Ri}[\not{\partial} + ie\not{A}]\psi_{Rj} \\ & -\sum_{ij}M_{ij}\bar{\psi}_{Li}\psi_{Rj} - \sum_{ij}M_{ij}^\dagger\bar{\psi}_{Ri}\psi_{Lj},\end{aligned}\tag{12.5.1}$$

where i, j are summed over the three lepton flavors (e , μ , and τ), ψ_{iL} and ψ_{iR} are the left- and right-handed parts of the field ψ_i , defined by

$$\psi_{Li} = \frac{1}{2}(\mathbf{1} + \gamma_5)\psi_i, \quad \psi_{Ri} = \frac{1}{2}(\mathbf{1} - \gamma_5)\psi_i,\tag{12.5.2}$$

and Z_L , Z_R , and M are numerical matrices. We are not assuming anything about lepton flavor conservation, so the matrices Z_{Lij} , Z_{Rij} and M_{ij} need not be diagonal. Also we are not assuming anything about invariance under P, C, or T invariance, so there is no necessary relation between Z_L and Z_R , or between M and $M^\dagger$. The only constraints on these matrices come from the reality of the Lagrangian density, which requires that Z_{Lij} and Z_{Rij} are Hermitian, and from the canonical anticommutation relations, which require that Z_{Lij} and Z_{Rij} are positive-definite.

⁵This section lies somewhat out of the book’s main line of development, and may be omitted in a first reading.

Now suppose we replace the lepton fields ψ_L, ψ_R with new fields ψ'_L, ψ'_R defined by

$$\psi_L = S_L \psi'_L, \quad \psi_R = S_R \psi'_R, \quad (12.5.3)$$

where $S_{L,R}$ are non-singular matrices that can be chosen as we like. The Lagrangian density when expressed in terms of these new fields then takes the same form as in Eq. (12.5.1), but with new matrices

$$Z'_L = S_L^\dagger Z_L S_L, \quad Z'_R = S_R^\dagger Z_R S_R, \quad M' = S_L^\dagger M S_R. \quad (12.5.4)$$

We can choose S_L and S_R so that $Z'_L = Z'_R = \mathbf{1}$. (Take $S_{L,R} = U_{L,R} D_{L,R}$, where $U_{L,R}$ are the unitary matrices that diagonalize the positive-definite Hermitian matrices $Z_{L,R}$, and the $D_{L,R}$ are the diagonal matrices whose elements are the inverse square roots of the eigenvalues of $Z_{L,R}$.)

Now make another transformation, to lepton fields ψ''_L defined by

$$\psi'_L = S'_L \psi''_L, \quad \psi'_R = S'_R \psi''_R. \quad (12.5.5)$$

The Lagrangian density again takes the same form when expressed in terms of these new fields, with new matrices

$$Z''_L = S'^{\dagger}_L S'_L, \quad Z''_R = S'^{\dagger}_R S'_R, \quad M'' = S'^{\dagger}_L M' S'_R. \quad (12.5.6)$$

This time we take $S'_{L,R}$ unitary, so that again $Z''_L = Z''_R = \mathbf{1}$. We choose these unitary matrices so that M'' is real and diagonal. (By the polar decomposition principle, M' like any square matrix may be put in the form $M' = V H$, where V is unitary and H is Hermitian. Take $S'_L = S'^{\dagger}_R V^\dagger$ and choose S'_R as the unitary matrix that diagonalizes H .) Dropping primes, the Lagrangian density now takes the form

$$\begin{aligned} \mathcal{L} = & -\frac{1}{4} Z_3 F_{\mu\nu} F^{\mu\nu} - \sum_i \bar{\psi}_{Li} [\not{\partial} + ie \not{A}] \psi_{Li} - \sum_i \bar{\psi}_{Ri} [\not{\partial} + ie \not{A}] \psi_{Ri} \\ & - \sum_i m_i \bar{\psi}_{Li} \psi_{Ri} - \sum_i m_i \bar{\psi}_{Ri} \psi_{Li}, \end{aligned} \quad (12.5.7)$$

where m_i are real numbers, the eigenvalues of the Hermitian matrix H . Finally, this can be put in the more familiar form

$$\mathcal{L} = -\frac{1}{4} Z_3 F_{\mu\nu} F^{\mu\nu} - \sum_i \bar{\psi}_i [\not{\partial} + ie \not{A}] \psi_i - \sum_i m_i \bar{\psi}_i \psi_i. \quad (12.5.8)$$

With the Lagrangian taking this form, it is now apparent that any renormalizable Lagrangian for lepton electrodynamics automatically conserves P, C, and T, as well as the numbers of leptons (minus the numbers of antileptons) of each flavor: electron, muon, and tauon.⁶ In particular, despite the appearance of Eq. (12.5.1), this theory does not allow

⁶This was first shown by Feinberg, Kabir, and myself [21]. Feinberg [22] had earlier noted that weak interaction effects in a theory with only one neutrino species would give rise to an observable rate for the process $\mu \rightarrow e + \gamma$, a difficulty that was only resolved by the discovery of a second neutrino species.

such processes as $\mu \rightarrow e + \gamma$. The reader may perhaps worry whether it is correct to identify the lepton fields as the ψ_i (previously called ψ_i'') appearing in Eq. (12.5.8), which obviously conserves lepton flavor, rather than the ψ_i appearing in Eq. (12.5.1), which seems to allow processes like $\mu \rightarrow e + \gamma$.

Such worries may be put aside; as stressed in Section 10.3, there is no one field that can be identified as *the* field of the electron or muon. In fact, although Eq. (12.5.1) yields a non-vanishing matrix element for radiative decay of lepton 1 into lepton 2 *off* the lepton mass shell, by taking the lepton momenta on the mass shell we find a vanishing S -matrix for all such processes even when calculated using Eq. (12.5.1).

It was essential in deriving these results that the same electric charge appeared in Eq. (12.5.1) for both the left- and right-handed parts of the lepton fields or, in other words, that both left- and right-handed parts of the lepton fields transform in the same way under electromagnetic gauge transformations. As we shall see in Volume II, for similar reasons the modern renormalizable theory of strong interactions known as quantum chromodynamics automatically conserves C, and (aside from certain non-perturbative effects) P and T, as well as the numbers of quarks (minus the numbers of antiquarks) of each quark flavor. We shall also see in Volume II that the simplest version of the renormalizable standard model of weak and electromagnetic interactions automatically conserves lepton flavor (though not C and P) for reasons similar to those described here for electrodynamics. It remains an open possibility that non-renormalizable interactions arising from higher mass scales may violate any of these conservation laws.

Problems

1. List all the renormalizable (or superrenormalizable) Lorentz-invariant terms in the Lagrangian of a single scalar field for space-time dimensionalities 2, 3, and 6.
2. Show how the overlapping divergence in the electron self-energy is cancelled in quantum electrodynamics.
3. Consider the theory of a scalar field ϕ and spinor field ψ , with interaction Hamiltonian $g\phi\bar{\psi}\psi$. Write the one-loop part of the scalar self-energy function $\Pi^*(q)$ as a divergent polynomial in p^μ , plus an explicit convergent integral.
4. Suppose that the quantum electrodynamics of electrons and photons is actually an effective field theory, derived by integrating out unknown particles of mass $M \gg m_e$. Assume gauge invariance and Lorentz invariance, but not invariance under C, P, or T. What are the non-renormalizable terms in the Lagrangian of leading order in $1/M$? Of next to leading order?

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